

Bhavani Adula

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SUMMARY

ML engineer working on **AI safety** and **model evaluation**: built the **adversarial evaluation** and **robustness testing** layer for a deployed **LLM** system, discovering and categorizing **300+ production failure modes** and hardening the model against them. Designs **research prototypes** with explicit experimental controls, including a **leakage-controlled** framework whose **negative result** shows fixed-threshold detectors miss real degradation. Modifies and trains **transformer** and **diffusion** architectures in **PyTorch**. MS Computer Science, May 2026.

TECHNICAL SKILLS

Deep Learning & Architectures: **PyTorch**, **transformers**, **sequence models**, **BERT**, **T5**, **Stable Diffusion** (UNet **cross-attention** modification), **LoRA / PEFT**, **Diffusers**, Hugging Face Transformers, XGBoost, SHAP

AI Safety & Evaluation: **adversarial testing**, **model evaluation**, **robustness testing**, **model validation**, **continuous monitoring**, **failure-mode analysis**, **root-cause analysis**, **eval datasets**, **regression testing**, safety guardrails, **calibration analysis**

Research Methods: experiment design, leakage controls, empirical analysis, statistical validation, baseline and ablation design, clustering (UMAP, HDBSCAN)

Languages & Engineering: **Python**, **SQL**, **R**, **Docker**, **CI/CD**, GitHub Actions, **pytest**, import-linter contracts, Git

Data & Pipelines: **large-scale multi-modal datasets** (video, sensor, text, audio), **data preprocessing**, **synthetic data generation**, PostgreSQL, pgvector, Airflow, pandas, NumPy

EDUCATION

University of Missouri-Kansas City — MS, Computer Science | GPA 3.7 May 2026

J.B. Institute of Technology — B.Tech, Information Technology | GPA 3.2 May 2023

EXPERIENCE

Hippocratic AI — AI/ML Engineer Intern | USA (Remote) Apr 2025 - May 2026

- Developed the **model evaluation** and **robustness testing** layer for a deployed **LLM** system: **240+ evaluation scenario cases** and **eval datasets**, with a **regression suite** giving **continuous monitoring** across releases that caught **8 regressions pre-release** and cut repeat failures in the largest category **40%**.
- Performed systematic **failure-mode discovery** on the deployed model, categorizing **300+ failed production conversations** through hallucination **root-cause analysis** into the taxonomy that drove every downstream fix.
- Hardened **robust inference** and production model control across **retrieval grounding**, **tool-calling** argument validation, and fallback and escalation logic: document-grounded accuracy up **12%**, invalid tool calls down **70%**, missed escalations down **45%**.
- Met production latency constraints on a resource-intensive model, cutting **p95 latency 25%** with no quality regression.

Yrall Media Solutions — Data Engineer | India Jan 2023 - Jun 2024

- Built scalable **Python/SQL** data pipelines on **PostgreSQL** and **Airflow** processing **20K-50K records per day**, with **data validation**, incremental loading, and automated monitoring.
- Wired a data-quality framework into **CI** to fail builds on regression, with record-level lineage tracing every output to its source.

PROJECTS

Silent Failure Detection — Label-Free Model Degradation Research Aug 2026

- Built a domain-agnostic research framework (**8,209 lines Python**, **261 tests**) testing whether unsupervised detectors catch model degradation without labels, backtesting a frozen model across **35 monthly windows** and **1,054,948 records**, with **import-linter** contracts enforcing a **leakage control** keeping the estimator away from the labels it is scored against.
- Negative result:** **AUC held flat (0.672 to 0.666)** while **Brier rose 17%** and the **calibration gap widened 27x**; of six unsupervised detectors three fired on **100% of windows** and one never fired, showing fixed-threshold alerting discards the trajectory signal that carries the degradation.

SS-SD — Synthetic Data Generation via Conditioned Diffusion Apr - May 2026

- Built a **synthetic data generation pipeline** for downstream perception-model training by modifying a pretrained diffusion architecture: replaced **Stable Diffusion 1.5's** CLIP text encoder with a custom **KinematicEncoder** projecting **76-dim robot kinematics** and gesture labels into the UNet's **cross-attention** conditioning space, fine-tuned with **LoRA** via **PEFT** on **JIGSAWS**.
- Handled **multi-modal data** (video, robot kinematics, gesture transcriptions, audio) and evaluated with **PSNR**, **SSIM**, **FID**, and **optical-flow temporal consistency**, plus a **validity control** scoring each generated frame against a different real frame to test whether the model tracks its conditioning signal or reproduces scene-average statistics.

ClinIQ — RAG Risk Detection | NSF NRT Research-A-Thon 2026, **4th Place** Apr - May 2026

- Clustered **3,316 flagged records** into **37 behavioral cohorts** with **UMAP** and **HDBSCAN** over 384-dim **embeddings** across **52,184 reviews (2008-2017)**, tracking risk signal rising **1.7% to 30.5%**, and authored the ethics and **auditability framework** grounding every model output in a cited government source.

Explainable Customer Churn Prediction Mar - May 2025

- Fine-tuned **BERT** for classification, reaching **0.793 accuracy** (macro F1 **0.73**) on **1,409 held-out customers**, chained **T5** for structured-to-text generation, and attributed drivers with **SHAP** over **XGBoost**.